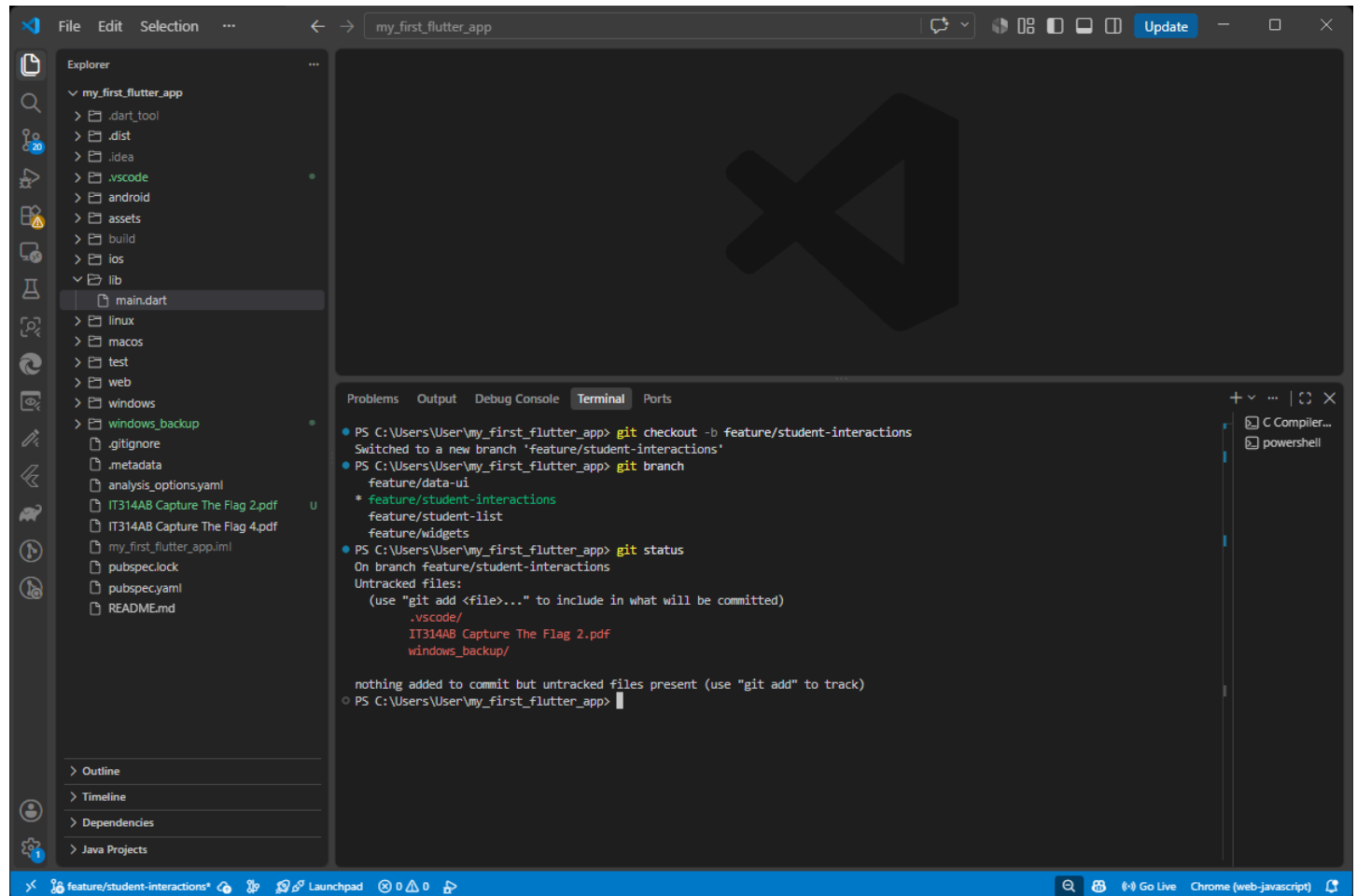


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Subject: IT314AB

Instructor: Canque Chiara Marie C.
Date: 08/20/2026

Capture the Flag 5: Make the System Interactive

Flag 0 — Create Your Feature Branch



The screenshot shows the Visual Studio Code interface with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'my_first_flutter_app' with various folders and files. The terminal is open to a PowerShell session in the directory 'C:\Users\User\my_first_flutter_app'. The terminal output shows the following commands and their results:

```
PS C:\Users\User\my_first_flutter_app> git checkout -b feature/student-interactions
Switched to a new branch 'feature/student-interactions'

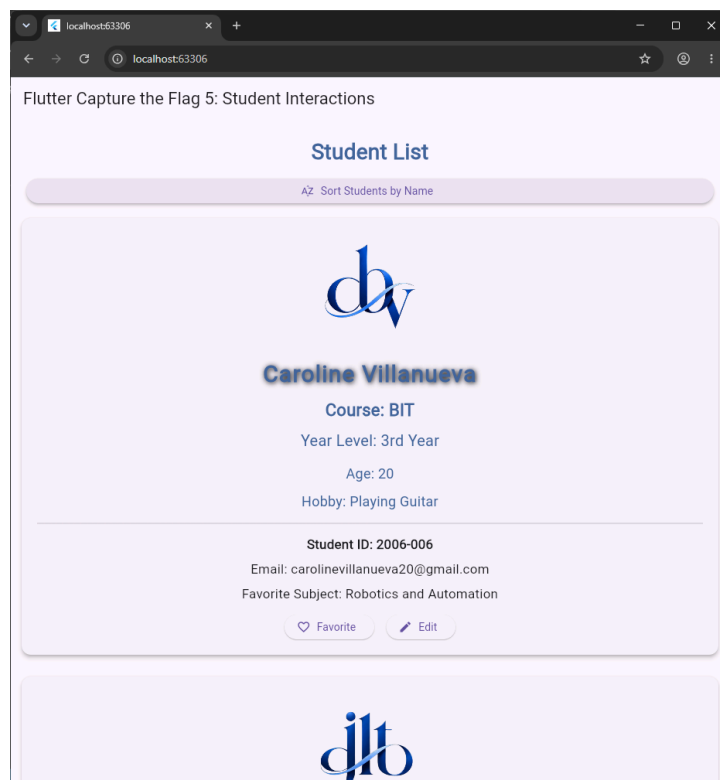
PS C:\Users\User\my_first_flutter_app> git branch
  feature/data-ui
* feature/student-interactions
  feature/student-list
  feature/widgets

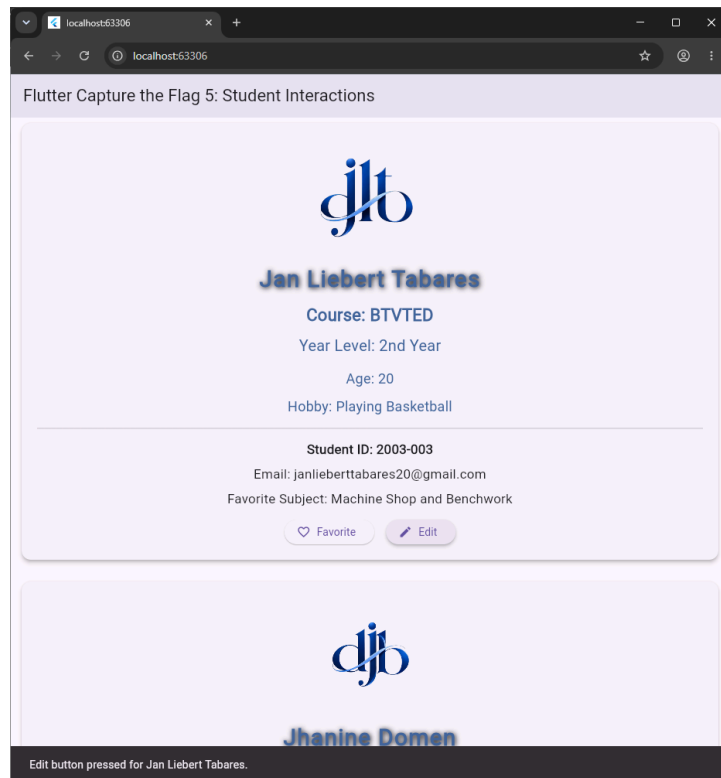
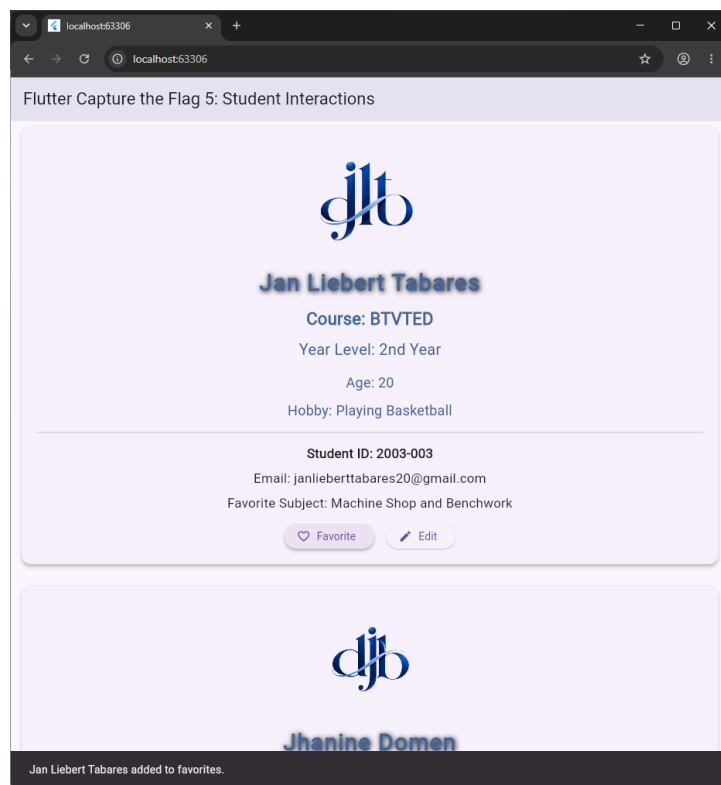
PS C:\Users\User\my_first_flutter_app> git status
On branch feature/student-interactions
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .vscode/
        IT314AB Capture The Flag 2.pdf
        windows_backup/

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\User\my_first_flutter_app>
```

Flag 1 — Wake Up the Buttons

```
lib > main.dart > ...
295 class ProfileCard extends StatelessWidget {
304   Widget build(BuildContext context) {
305     const SizedBox(height: 20),
306     Row(
307       mainAxisAlignment: MainAxisAlignment.center,
308       children: [
309         ElevatedButton.icon(
310           onPressed: () {
311             ScaffoldMessenger.of(context).showSnackBar(
312               SnackBar(
313                 content: Text(
314                   '${profile.name} added to favorites.',
315                 ), // Text
316               ), // SnackBar
317             );
318           },
319           icon: const Icon(Icons.favorite_border),
320           label: const Text('Favorite'),
321         ), // ElevatedButton.icon
322         const SizedBox(width: 15),
323         ElevatedButton.icon(
324           onPressed: () {
325             ScaffoldMessenger.of(context).showSnackBar(
326               SnackBar(
327                 content: Text(
328                   'Edit button pressed for ${profile.name}.',
329                 ), // Text
330               ), // SnackBar
331             );
332           },
333           icon: const Icon(Icons.edit),
334           label: const Text('Edit'),
335         ), // ElevatedButton.icon
336       ], // Row
337     ), // Column
338   ), // Padding
339 ); // Card
340 }
```

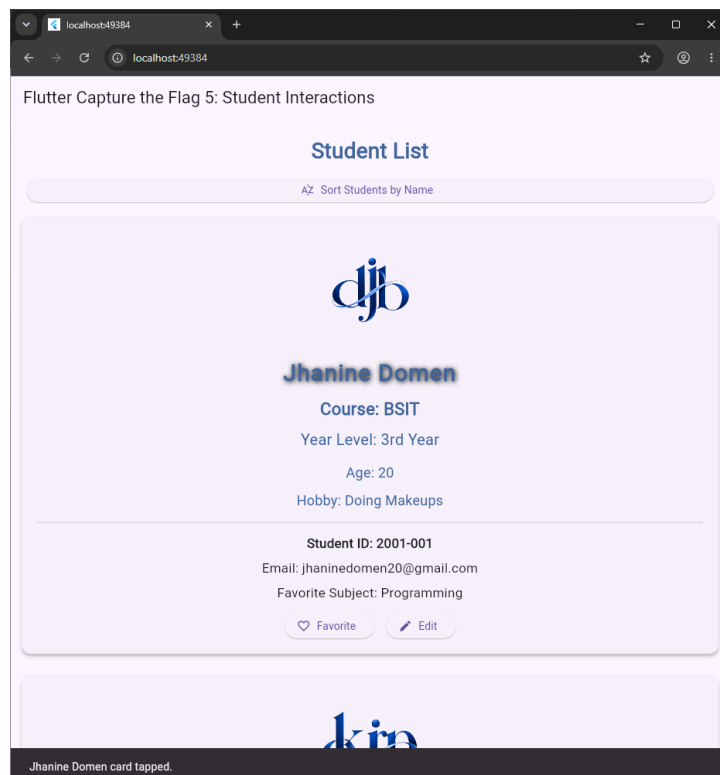


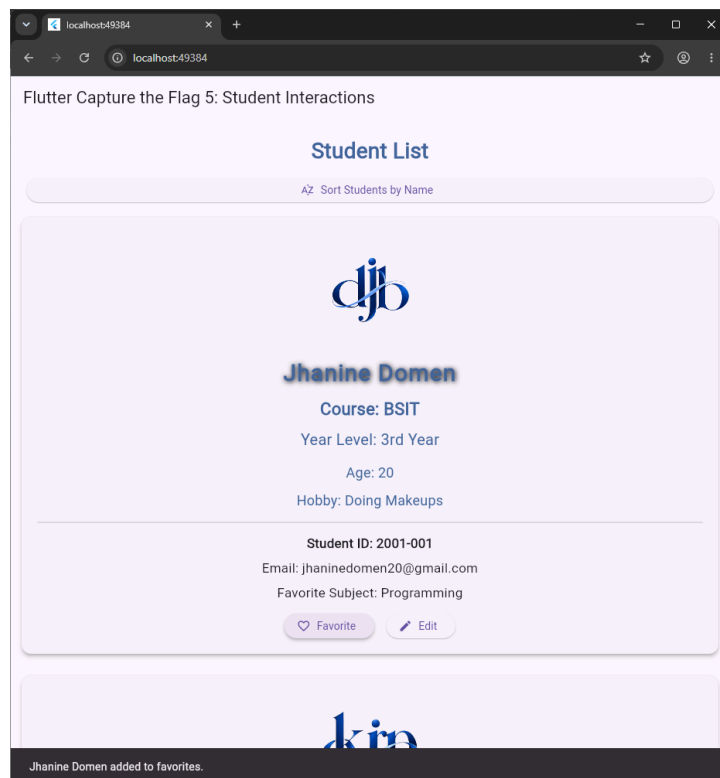
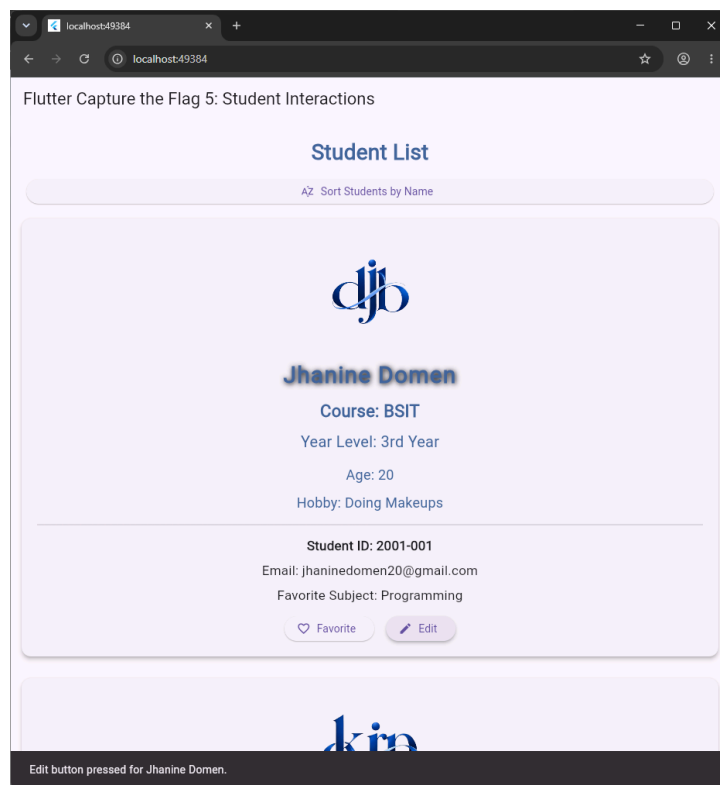


Flag 2 — Touch Anywhere

```
File Edit Selection View Go ... my_first_flutter_app

lib > main.dart > ProfileCard > build
261 class EmptyStudentState extends StatelessWidget {
262   // ...
263 }
264
265 class ProfileCard extends StatelessWidget {
266   final Profile profile;
267
268   const ProfileCard({
269     super.key,
270     required this.profile,
271   });
272
273   @override
274   Widget build(BuildContext context) {
275     return GestureDetector(
276       onTap: () {
277         ScaffoldMessenger.of(context).showSnackBar(
278           SnackBar(
279             content: Text(
280               '${profile.name} card tapped.',
281             ), // Text
282           ), // SnackBar
283         );
284       },
285       child: Card(
286         elevation: 4,
287         child: Padding(
288           padding: const EdgeInsets.all(20),
289           child: Column(
290             children: [
291               Image.asset(
292                 profile.image,
293                 width: 150,
294                 height: 150,
295                 fit: BoxFit.cover,
296               ), // Image.asset
297               const SizedBox(height: 10),
298               Text(
299                 profile.name,
300                 textAlign: TextAlign.center,
301               ),
302             ],
303           ),
304         ),
305       ),
306     );
307   }
308 }
```

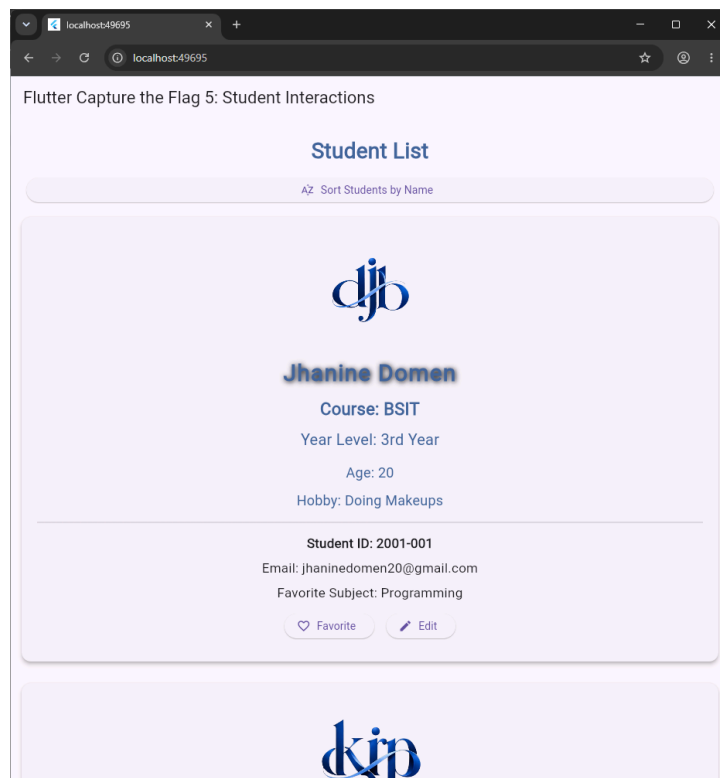


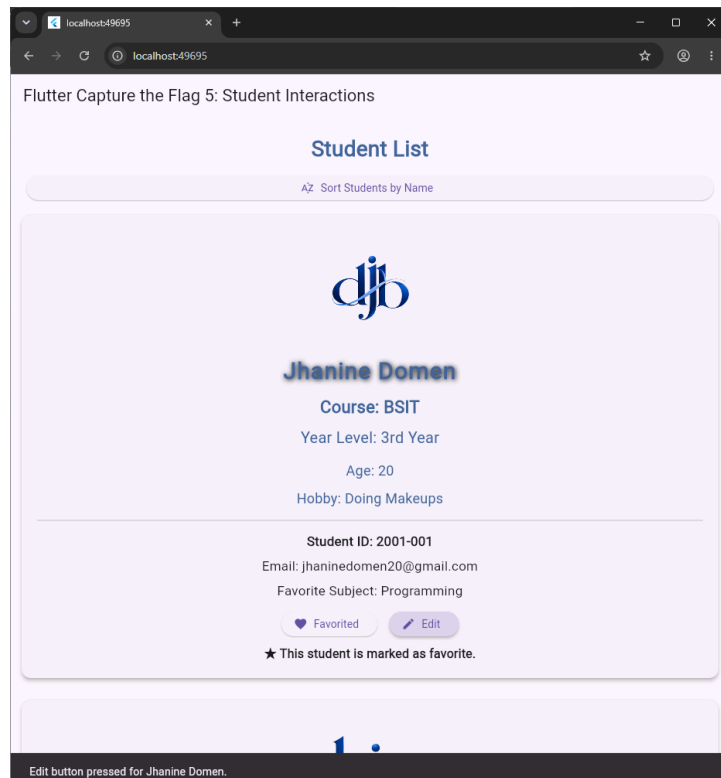
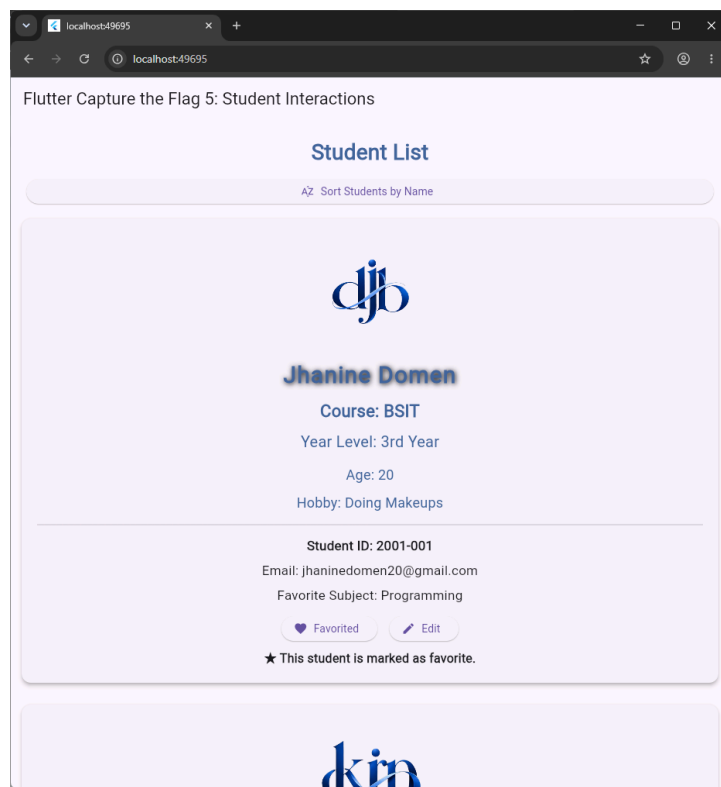


Flag 3 — Discover **setState**

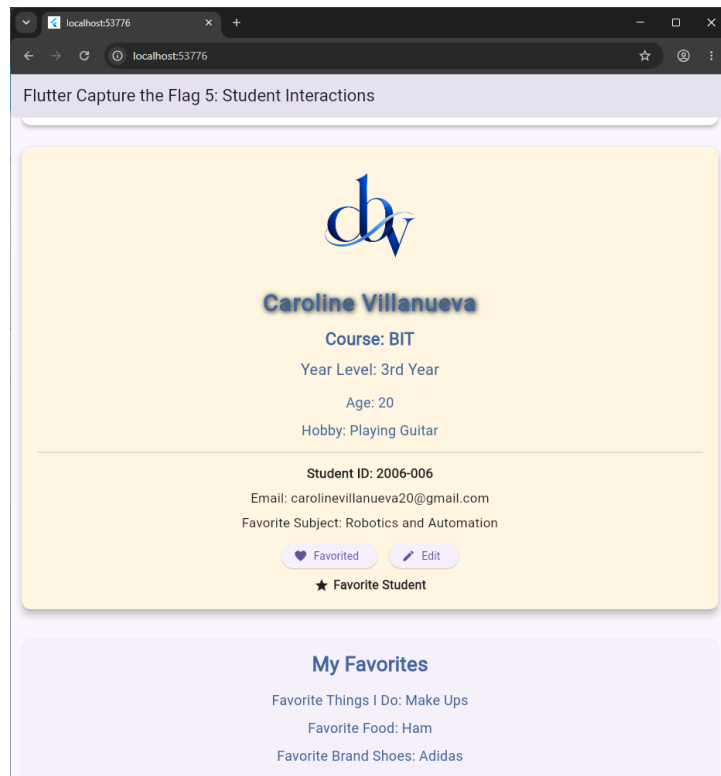
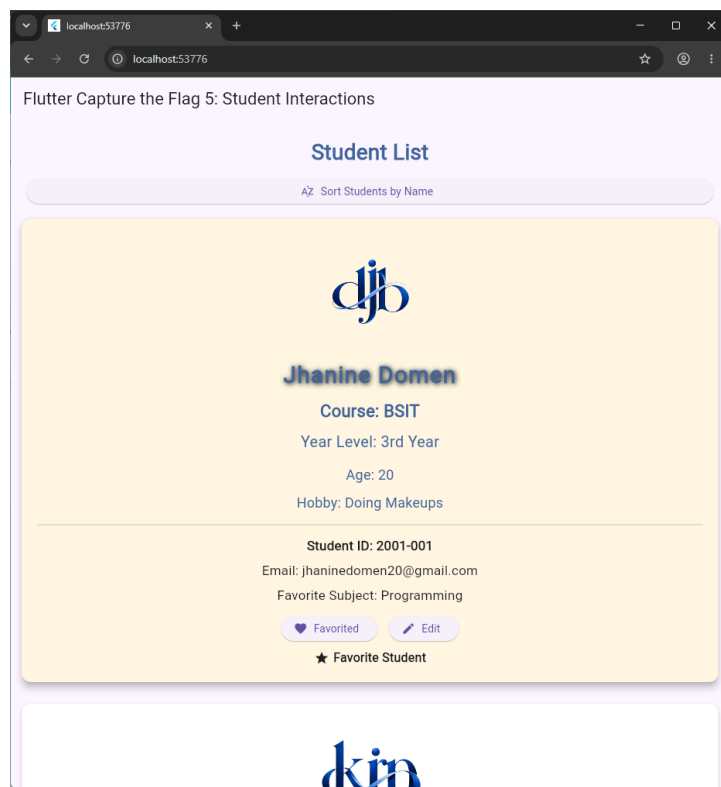
```
File Edit Selection ... my_first_flutter_app Update
Explorer
my_first_flutter_app
  .dart_tool
  .dist
  .idea
  .vscode
  android
  assets
  build
  ios
  lib
    main.dart
  linux
  macos
  test
  web
  windows
  windows_backup
  .gitignore
  .metadata
  analysis_options.yaml
  IT314AB Capture The Flag 2.pdf
  IT314AB Capture The Flag 4.pdf
  my_first_flutter_app.iml
  pubspec.lock
  pubspec.yaml
  README.md
  Outline
  Timeline
  Dependencies
  Java Projects

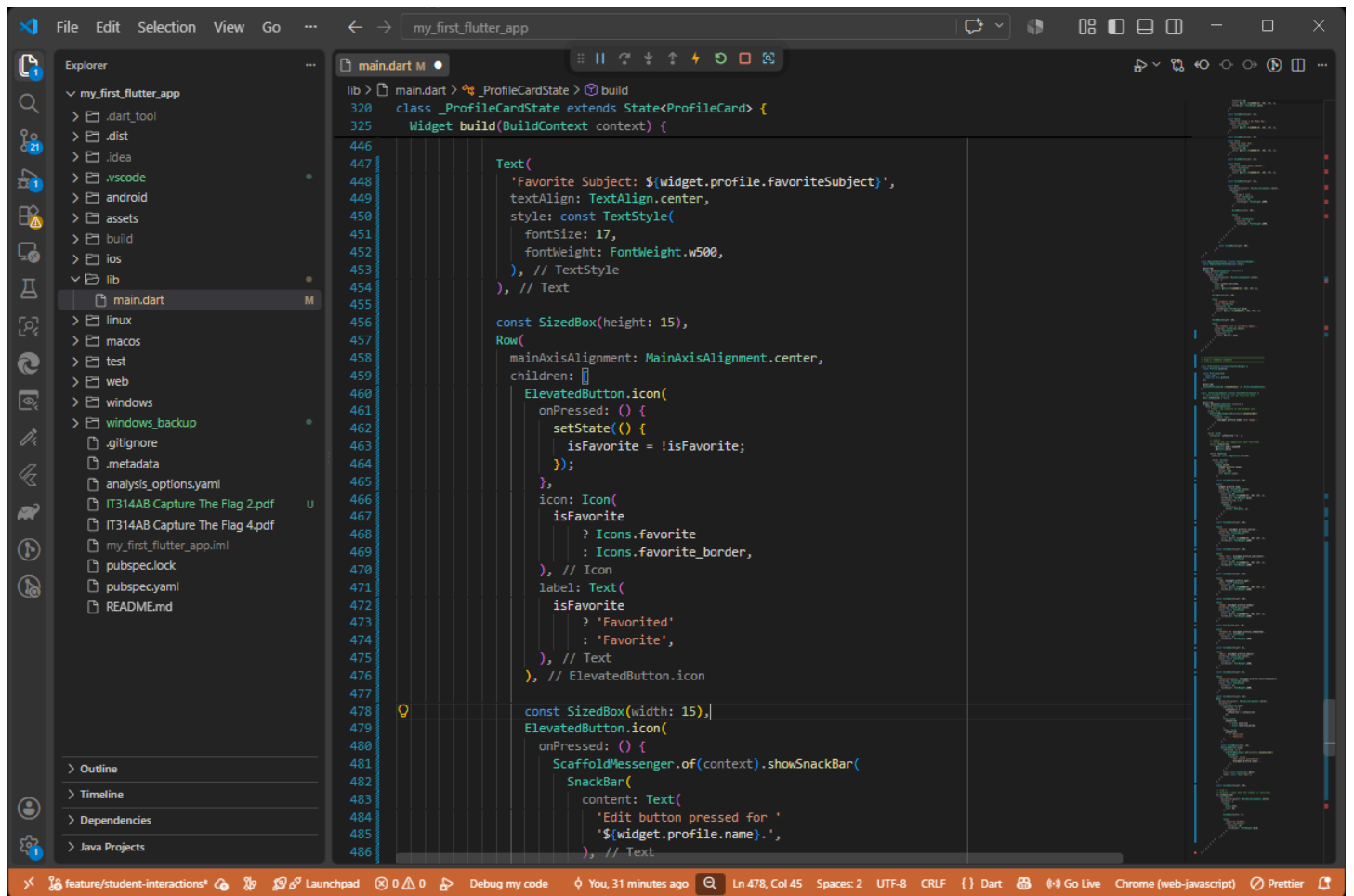
main.dart
320 class _ProfileCardState extends State<ProfileCard> {
325   Widget build(BuildContext context) {
450
451     Row(
452       mainAxisAlignment: MainAxisAlignment.center,
453       children: [
454         ElevatedButton.icon(
455           onPressed: () {
456             setState(() {
457               isFavorite = !isFavorite;
458             });
459           },
460           icon: Icon(
461             isFavorite
462               ? Icons.favorite
463               : Icons.favorite_border,
464           ), // Icon
465
466           label: Text(
467             isFavorite ? 'Favorited' : 'Favorite',
468           ), // Text
469         ), // ElevatedButton.icon
470
471         const SizedBox(width: 15),
472         ElevatedButton.icon(
473           onPressed: () {
474             ScaffoldMessenger.of(context).showSnackBar(
475               SnackBar(
476                 content: Text(
477                   'Edit button pressed for '
478                   '${widget.profile.name}.',
479                 ), // Text
480               ), // SnackBar
481             );
482           },
483           icon: const Icon(Icons.edit),
484           label: const Text('Edit'),
485         ), // ElevatedButton.icon
486       ], // Row
487     ),
488
489     const SizedBox(height: 10),
490     if (isFavorite)
491       const Text(
```





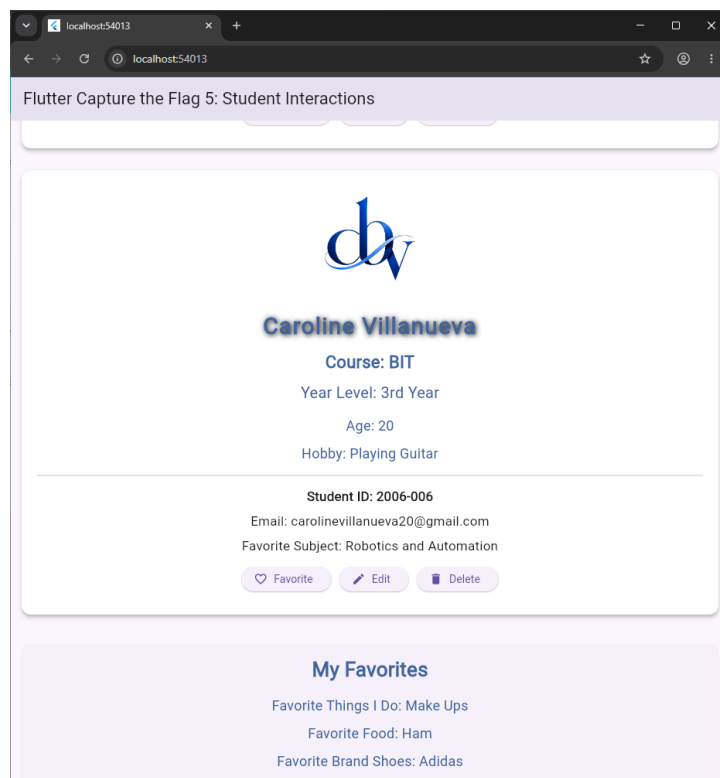
Flag 4 – Favorite Students





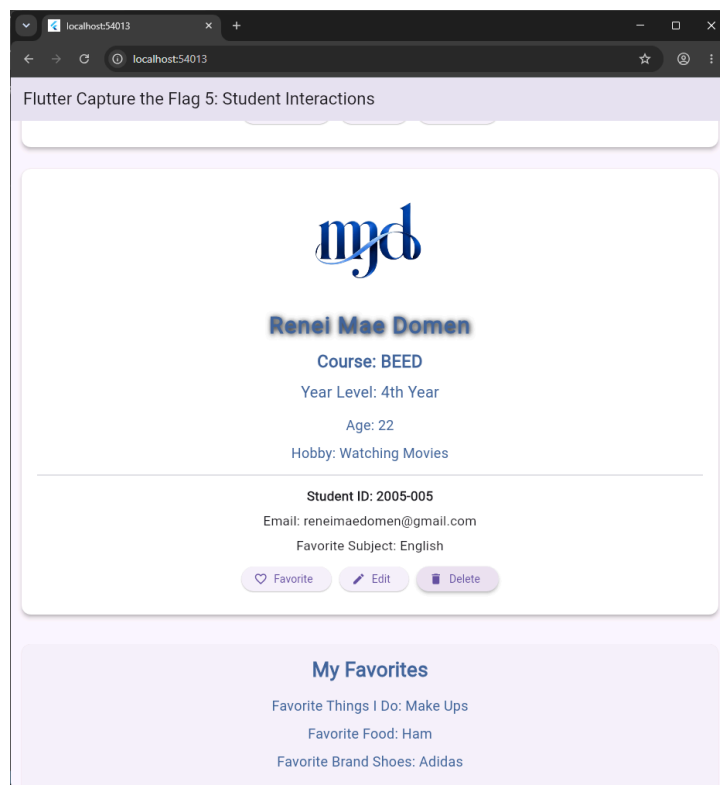
```
lib > main.dart > _ProfileCardState > build
320 class _ProfileCardState extends State<ProfileCard> {
325   Widget build(BuildContext context) {
446
447     Text(
448       'Favorite Subject: ${widget.profile.favoriteSubject}',
449       textAlign: TextAlign.center,
450       style: const TextStyle(
451         fontSize: 17,
452         fontWeight: FontWeight.w500,
453       ), // TextStyle
454     ), // Text
455
456     const SizedBox(height: 15),
457     Row(
458       mainAxisAlignment: MainAxisAlignment.center,
459       children: [
460         ElevatedButton.icon(
461           onPressed: () {
462             setState(() {
463               isFavorite = !isFavorite;
464             });
465           },
466           icon: Icon(
467             isFavorite
468               ? Icons.favorite
469               : Icons.favorite_border,
470           ), // Icon
471           label: Text(
472             isFavorite
473               ? 'Favorited'
474               : 'Favorite',
475           ), // Text
476         ), // ElevatedButton.icon
477
478         const SizedBox(width: 15),
479         ElevatedButton.icon(
480           onPressed: () {
481             ScaffoldMessenger.of(context).showSnackBar(
482               SnackBar(
483                 content: Text(
484                   'Edit button pressed for '
485                   '${widget.profile.name}',
486                   // Text
487                 ),
488               ),
489             );
490           },
491           icon: Icon(Icons.edit),
492           label: Text('Edit'),
493         ), // ElevatedButton.icon
494       ],
495     ), // Row
496   ), // build
497 }
```

Flag 5 — Remove a Student

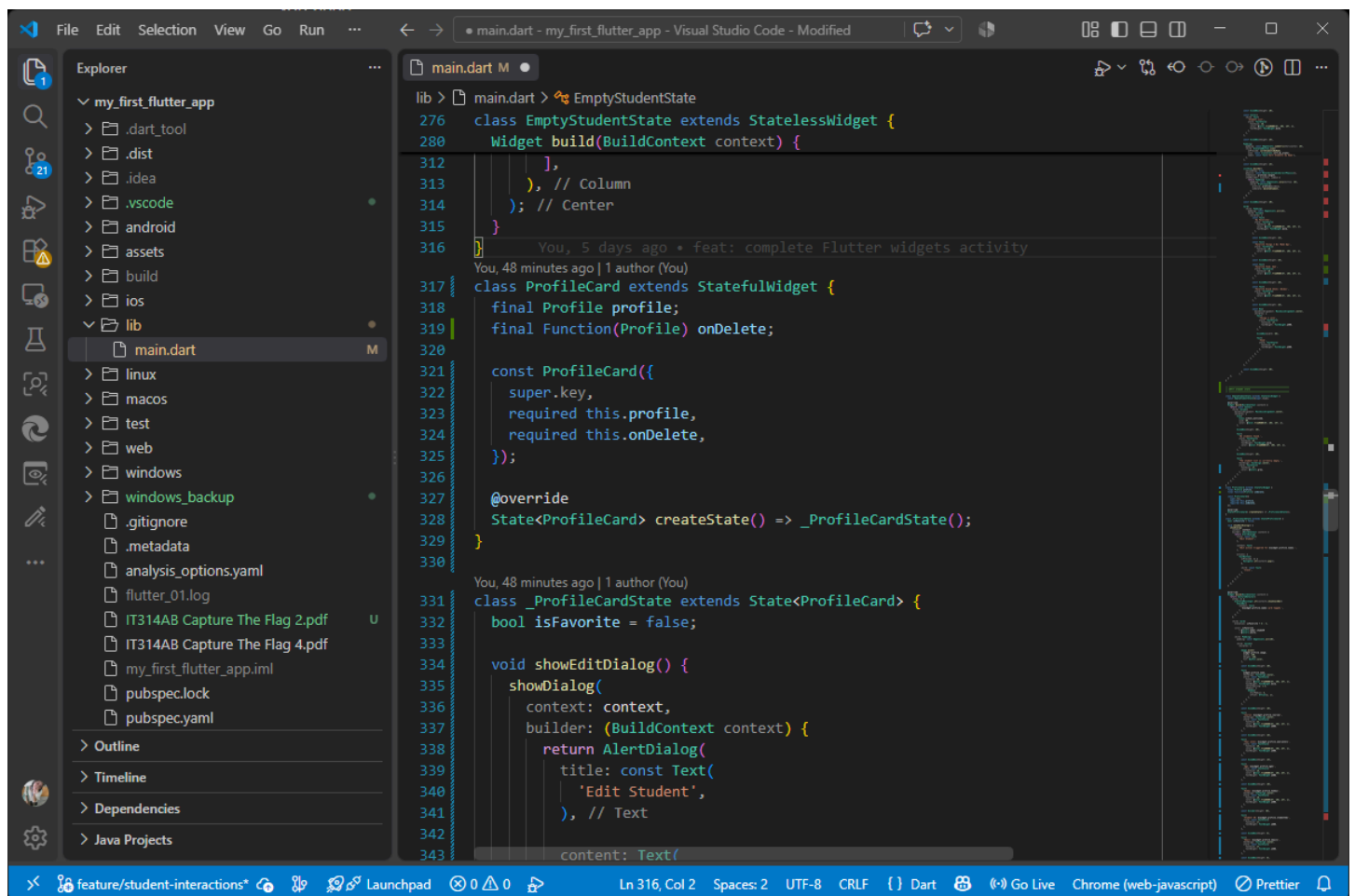


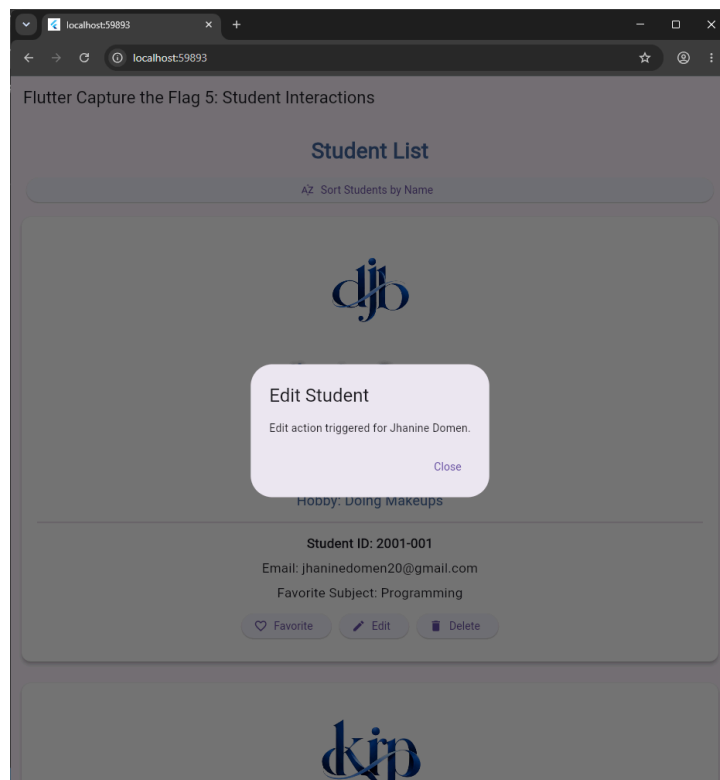
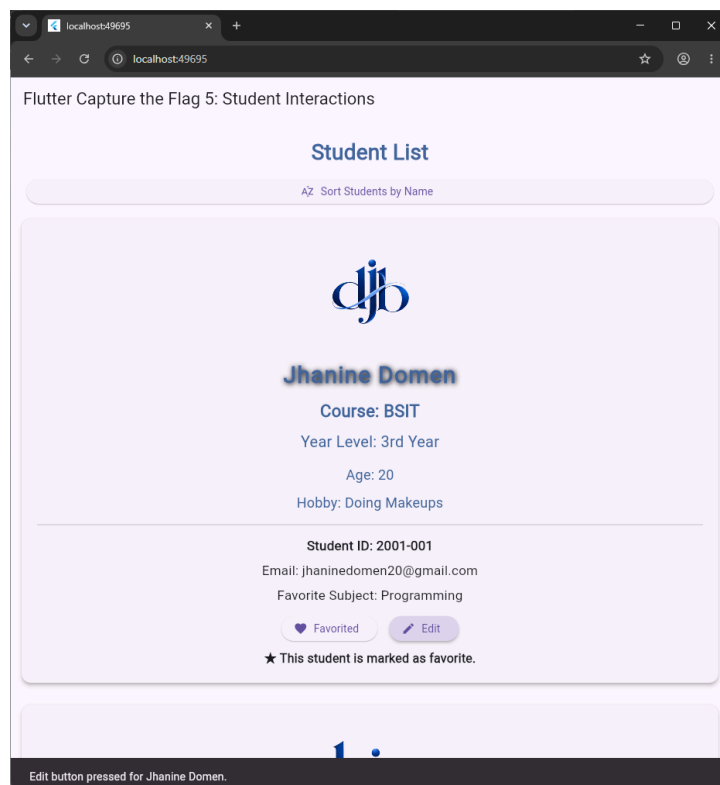
```
lib > main.dart > _MyHomePageState
112 class MyHomePage extends StatefulWidget {
113   StatefulElement createElement() => _MyHomePageState(),
114 }
115
You, 13 hours ago | 1 author (You)
119 class _MyHomePageState extends State<MyHomePage> {
120   void sortStudentsByName() {
121     setState(() {
122       profiles.sort((a, b) => a.name.compareTo(b.name));
123     });
124   }
125
126   void deleteStudent(Profile profile) {
127     setState(() {
128       profiles.remove(profile);
129     });
130   }
131
132   @override
133   Widget build(BuildContext context) {
134     return Scaffold(
135       appBar: AppBar(
136         title: const Text(
137           'Flutter Capture the Flag 5: Student Interactions',
138         ), // Text
139       ), // AppBar
140       body: profiles.isEmpty
141         ? const EmptyStudentState()
142         : ListView(
143           padding: const EdgeInsets.all(10),
144           children: [
145             const SizedBox(height: 10),
146             const Center(
147               child: Text(
148                 'Student List',
149
```

```
lib > main.dart > _ProfileCardState > build
268 class EmptyStudentState extends StatelessWidget {
272   Widget build(BuildContext context) {
305     ), // Column
306   ); // Center
307 }
308 }
309
...
310 class ProfileCard extends StatefulWidget {
311   final Profile profile;
312
313   final Function(Profile) onDelete;
314
315   const ProfileCard({
316     super.key,
317     required this.profile,
318     required this.onDelete,
319   });
320
321   @override
322   State<ProfileCard> createState() => _ProfileCardState();
323 }
324
...
325 class _ProfileCardState extends State<ProfileCard> {
326   bool isFavorite = false;
327
328   @override
329   Widget build(BuildContext context) {
330     return GestureDetector(
331       onTap: () {
332         ScaffoldMessenger.of(context).showSnackBar(
333           SnackBar(
334             content: Text(
335               '${widget.profile.name} card tapped.',
336
```

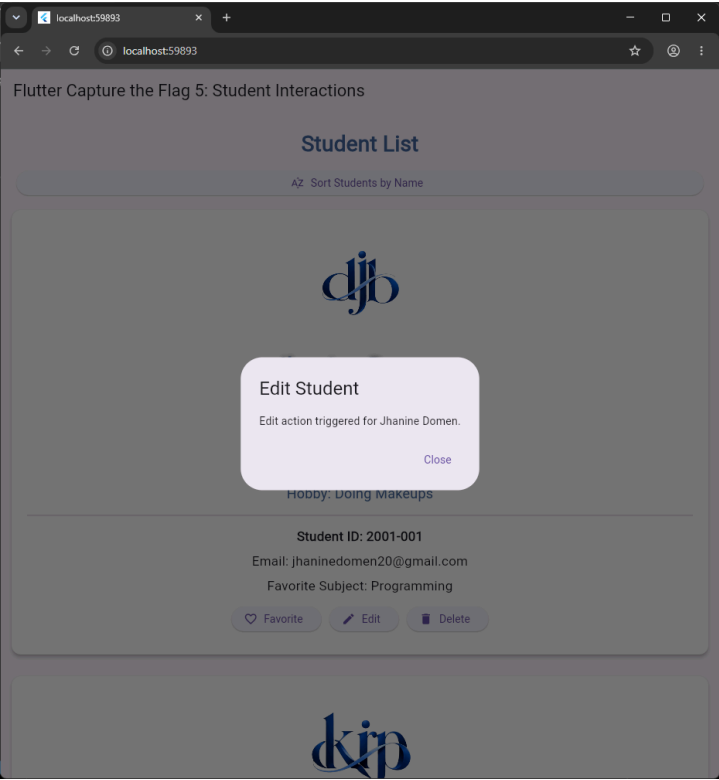
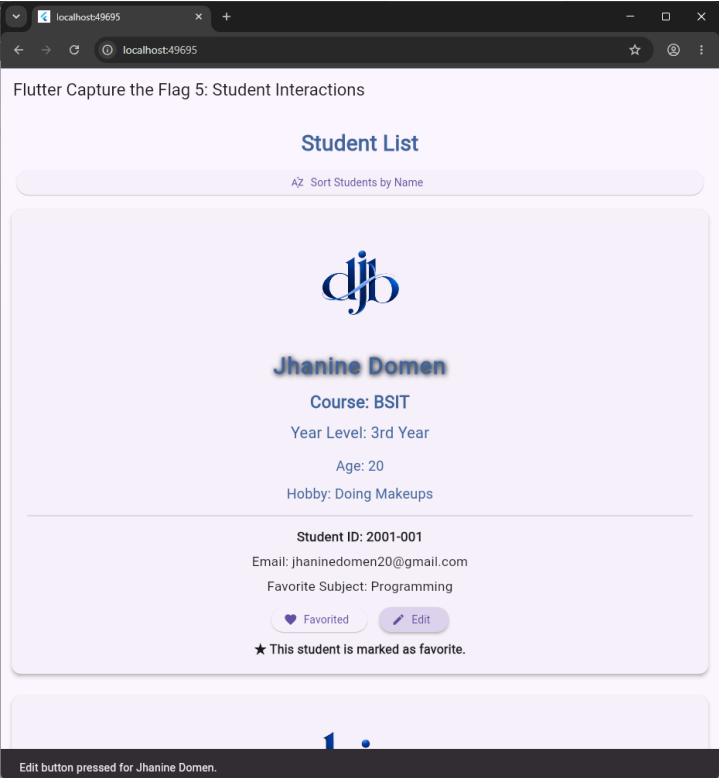


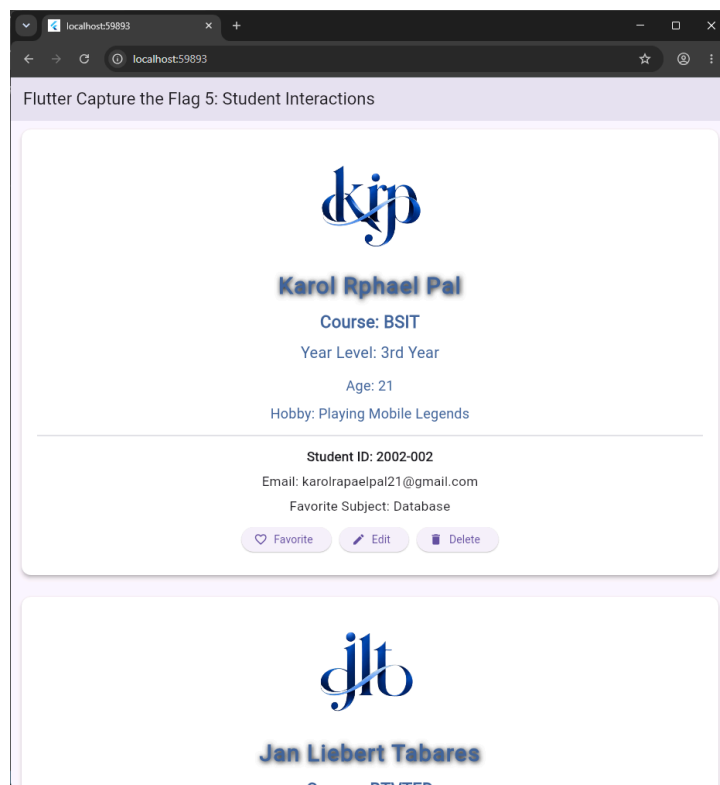
Flag 6 — Edit Trigger





Flag 7 — Multiple Actions, Independent State





Flag 8 — Save Your Progress

```
MINGW64:/c/Users/User/my_first_flutter_app
To https://github.com/DomenJhanineB/IT314AB_Domen.git
* [new branch]      feature/student-interactions -> feature/student-interactions
s

User@domenjhanineb MINGW64 ~/my_first_flutter_app (feature/student-interactions)
$ git status
On branch feature/student-interactions
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .vscode/
    IT314AB Capture The Flag 2.pdf
    windows_backup/

nothing added to commit but untracked files present (use "git add" to track)

User@domenjhanineb MINGW64 ~/my_first_flutter_app (feature/student-interactions)
$ git branch
  feature/data-ui
* feature/student-interactions
  feature/student-list
  feature/widgets

User@domenjhanineb MINGW64 ~/my_first_flutter_app (feature/student-interactions)
$
```